



Board of Commissioners
Thursday, February 13, 2025
2:00 p.m.



Board of Commissioners
Central Arkansas Water
AGENDA

2:00 PM

Thursday, February 13, 2025

James T. Harvey Administration Building

221 East Capitol Avenue

Little Rock, Arkansas

*CAW board meetings are open to in-person attendance by the public
unless a Public Health Emergency is declared in Pulaski County.
CAW will continue to provide a means for the public to attend online.*

1. *You can view the meeting electronically by clicking [here](#).*
2. *The public may submit written comments and questions or request an opportunity to address the Board of Commissioners for up to 3 minutes on any subject relevant to CAW during the public comment period by emailing PublicComments@carkw.com no later than 11:00am prior to the meeting. If you request to speak virtually, when your name is called by the Chair, you should "raise your hand" in Microsoft Teams and then your microphone will be unmuted. Written comments and questions will be provided to the Commission before the meeting.*

Page

1) CALL TO ORDER (Chair Smith)

2) APPROVAL OF MINUTES (Chair Smith)

a) [January 16, 2025, Regular Meeting](#) 

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3) PUBLIC COMMENT PERIOD (Chair Smith)

4) FINANCIAL REPORT

(Cynthia Edwards, Director of Finance)

5) INFORMATION SHARING (Chair Smith)

- a) Jack H. Wilson WTP Renewal & Resiliency Project Update
(Andrew Pownall, Senior Engineer)

6) CHIEF EXECUTIVE'S REPORT

(Tad Bohannon, Chief Executive Officer)

7) CONSENT AGENDA (Chair Smith)

- a) USGS Contract for the 2025 Streamflow and Water Quality Monitoring Work Plan 9
(Bradley Meredith, Water Quality Specialist)
Staff requests Board approval to renew the annual contract with the USGS to monitor streamflows and water quality at Lake Maumelle and Lake Winona. USGS streamflow and monitoring work at the lakes has been ongoing since 1989. The total cost of the contract is \$449,455.00; CAW's direct cost is \$368,415.00.
[Agenda Item](#) 
- b) Lake Maumelle Bathymetry 10 - 15
(Alex Harper, GIS Manager)
Staff requests Board approval to exercise the second of three one-year extensions to the existing contract with Sanborn in the amount of \$198,704.47 to complete the bathymetric mapping of Lake Maumelle.
[Agenda Item](#) 
[Quote](#) 
- c) 2-Inch Galvanized Water Main Replacements in Little Rock - 16 - 18
2025 - Phase 2
(Jim Ferguson, Director of Engineering)
Staff requests Board approval of a construction contract with Cisneros Family Construction in the amount of \$1,893,499.26 to replace approximately 7,410 linear feet of existing water mains identified as being very problematic and high maintenance. The area of this project is near Boyle Park and Rock Creek Golf Course and areas located south of Colonel Glenn Road and west of University Avenue in Little Rock.
[Agenda Item](#) 
[Map](#) 
[Bid Sheet](#) 

8) NON-CONSENT AGENDA (Chair Smith)

a) Water System Optimization Analysis

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(Andrew Pownall, Senior Engineer)

Staff requests Board approval for an engineering services agreement with Arcadis in the amount of \$378,000. This agreement will support a study designed to guide operational optimization in our distribution system. The analysis will utilize the recently updated hydraulic model to provide a comprehensive, detailed evaluation of operating scenarios for tanks and pump stations and will make recommendations to enhance both water quality and energy efficiency.

[Agenda Item](#) 

9) ADJOURN (Chair Smith)

10) DEPARTMENT REPORTS

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(for informational purposes only)

1. Finance
2. [Customer Service](#) 
3. [Engineering](#) 
4. [Distribution](#) 
5. [Water Production](#) 
6. [Watershed](#) 

Minutes of a Regular Meeting
Board of Commissioners
Central Arkansas Water

January 16, 2025

The Board of Commissioners of Central Arkansas Water (CAW) convened in a regular meeting at 2:00 p.m. on Thursday, January 16, 2025, via Teams and in the Board Room of the James T. Harvey Administration Building located at 221 East Capitol Avenue, Little Rock, Arkansas.

Present:

Ms. Carmen Smith, Chair
Mr. Jay Hartman, Vice Chair
Mr. Kevin Newton, Secretary-Treasurer
Ms. Kandi Hughes, Commissioner
Mr. Jim McKenzie, Commissioner
Dr. Jay Barth, Commissioner
Ms. Melanie Hoggard, Commissioner

CAW Staff in Attendance:

Mr. Tad Bohannon, Chief Executive Officer
Mr. Blake Weindorf, P.E., Chief Operating Officer
Mr. David Johnson, General Counsel
Ms. Jo Courtney, Chief Administrative Officer
Ms. Cynthia Edwards, Director of Finance
Ms. Tina Richardson, Administrative Assistant - Accounting
Mr. Danny Dunn, Director of Distribution
Ms. Teletha Leonard, Director of Human Resources
Mr. Allen Vincent, Director of Information Services
Mr. Alex Harper, GIS Manager
Mr. Sam Zethaban, Director of Water Production
Mr. Brad Meredith, Water Quality Specialist
Mr. Jim Ferguson, P.E., Director of Engineering
Ms. Elizabeth Heiles, P.E., Senior Engineer
Mr. Doug Shackelford, Public Affairs Officer
Ms. Heide Harrell, Director of Communications

CAW Staff via Teleconferencing:

Mr. Andrew Pownall, P.E., Senior Engineer
Ms. Angeline Baldwin, Senior Accountant
Mr. Benjamin Thesing, Lead Natural Resource Specialist
Mr. Bryan Rugar, Land Conservation Coordinator
Ms. Danielle Nielsen, Accounting Clerk II
Mr. Doug Graham, Optimization Manager
Mr. Itayi Pondwa, Finance Manager
Ms. Jane Hurley, Education and Outreach Specialist
Mr. Jason Lowder, Engineering Technician
Mr. Joe Ginn, Information Services Technician I
Mr. Jordan Qualls, Engineering Technician
Ms. Karen Baker, HRIS & Benefits Administrator

Ms. Liz Tuck-Rowan, Purchasing Manager
Mr. Patrick Greene, Customer Service Field Manager
Ms. Rebecca Mitchell, Training & Development Specialist
Ms. Samantha Williams-Davis, Communications Specialist & Brand Manager
Ms. Sarah Howard, Human Resources Manager
Ms. Suzette Ginn, Communications Assistant
Mr. Terry Frazier, Director of Customer Service

Also, in Attendance:

Mr. Bailey Moll, JPJ
Mr. Paul Strickland, P.E., Garver
Ms. Carolyn Sherman, Friends of Ouachita Trail
Mr. Chuck Soloman, American Pipe

CALL TO ORDER

Chair Smith called the meeting to order at 2:00 p.m.

SERVICE AWARD PRESENTATION

Mr. Bohannon recognized Ms. Courtney for one year of dedicated service. Ms. Edwards recognized Ms. Richardson for 45 years of dedicated service.

APPROVAL OF BOARD MINUTES

The minutes of the December 12, 2024, regular meeting were unanimously approved upon a motion duly made by Commissioner Hartman and seconded by Commissioner McKenzie

FINANCIAL REPORT

Ms. Edwards provided a year-to-date financial update on utility operations, the Semi-Annual Investment Report, and an APERS update.

INFORMATION SHARING

USGS Water Quality Monitoring on CAW Lakes

Mr. Meridith provided an update on the cooperative agreement between CAW and the US Geological Survey (USGS) monitoring program. He stated that the partnership has been in place since 1989, and it assists with monitoring locations and providing real-time data, which helps Treatment Facility Operators quickly adjust to changing water conditions. The 2025 USGS Work Plan will be on the Consent Agenda in February for Board approval.

CHIEF EXECUTIVE'S REPORT

Mr. Bohannon provided an update on several matters to the Board.

CONSENT AGENDA

Annual Software Maintenance for CAW Computerized Maintenance Management System

Staff requested Board approval of an annual software maintenance contract in the amount of \$128,500.00 with Cityworks.

Annual Contract for Asphalt Surface Repairs

Staff requested Board approval of a contract with Redstone Construction Group, estimated at \$830,045.00, to perform asphalt restoration work needed after CAW crews make repairs or modifications to existing mains. Actual payment will be based on completed work and unit costs.

Purchase Crew and Dump Trucks

Staff requested Board approval to purchase two (2) dump trucks with dump beds and two (2) crew trucks with service bodies. The total purchase price for all four units is \$616,682.00.

Authorization to Condemn Easements for a Force Main, Outfall, and Wastewater for the Wastewater System in Wrightsville

Staff requested Board approval to pursue the condemnation of easements to install, construct, house, contain, maintain, repair, and eventually replace a force main and outfall of the wastewater system that serves the Wrightsville community and other customers and to pursue the condemnation of an easement to dispose of, drain, and move wastewater from the force main and outfall on real property owned by the Randy W. Thomas Living Trust.

The Consent Agenda was unanimously approved upon a motion duly made by Commissioner McKenzie and seconded by Commissioner Hartman.

NON-CONSENT AGENDA

Engineering Services for 60-Inch Raw Water Line Route Study

Ms. Heiles requested Board approval of an engineering services contract in the amount of \$545,000.00 with Garver Engineers, Inc., to perform a route study for a new 60-inch raw water line from Lake Maumelle to the Jack H. Wilson Water Treatment Plant (WTP). Ms. Heiles stated CAW currently operates a 48-inch line to the Jackson Reservoir and a 72-inch to the WTP. Staff are asking the Board to consider a third line because there is no redundancy with the 72-inch line. CAW did a condition assessment in 2024, and while the lines are in good shape, there is a need to plan, especially with increasing capacity at WTP. The study helps identify property acquisitions, evaluates the system hydraulics and environmental permitting impacts, and includes 10% construction drawings, allowing for budgeting and projected timelines. Ms. Heiles presented a task timeline for a best-case scenario.

Commissioner Barth asked how many of these tasks are covered by the current rate plan. Mr. Weindorf stated that the route study, design, and construction of the 60" line is included in the 10-year rate model as part of the infrastructure fee.

After consideration and discussion, the contract was unanimously approved upon a motion duly made by Commissioner Barth and seconded by Commissioner Hoggard.

Lake Maumelle and Jackson Reservoir Dam Repairs

Mr. Ferguson requested Board approval of a construction contract in the amount of \$1,290,803.41 with James A. Rogers Excavating to repair the upstream slope of Lake Maumelle Dam and an immediate change order in the amount of \$300,000 to perform the dam slope repair at Jackson

Reservoir. Mr. Ferguson described the current damage to the Lake Maumelle and Jackson Reservoir dams.

After consideration and discussion, the contract and change order were unanimously approved upon a motion duly made by Commissioner McKenzie and seconded by Commissioner Barth.

Classification and Compensation Study

Ms. Leonard requested Board approval of a contract with Segal Company for \$145,000.00 to perform a Classification and Compensation Study of 140 titles and an additional statement of work with Segal Company for an Annual Merit Bonus System and Performance Evaluation Review for \$50,000.

Commissioner Barth asked about the expected timeline for completion. Ms. Leonard said the entire process would take about six months.

Commissioner Hartman asked what is expected from the \$10,000 final results presentation. Ms. Leonard said Segal would meet with HR to review their results and recommendations and work with HR as needed to prepare for the board presentation. There will also be two days for on-site meetings with employees.

After consideration and discussion, the contract was unanimously approved upon a motion duly made by Commissioner McKenzie and seconded by Commissioner Hartman.

EXECUTIVE SESSION

The Board went into Executive Session to discuss the annual review of the General Counsel and CEO upon a motion duly made by Commissioner Barth and seconded by Commissioner Hughes.

Upon reconvening the Regular Session, upon a motion duly made by Commissioner Barth and seconded by Commissioner Hughes, the Board unanimously approved an increase in the CEO's salary by 3.5%, which is the same COLA raise given to other CAW employees, an additional 3% increase for Merit Pay, and a \$10,000.00 bonus.

ADJOURNMENT

With no further business, Chair Smith adjourned the meeting.

Kevin Newton, Secretary-Treasurer

DATE _____

AMT _____

CENTRAL ARKANSAS WATER

BOARD OF COMMISSIONERS

AGENDA ITEM

Date: February 13, 2025 **Department:** Water Production **Project ID#:** n/a

Item Title: 2025 USGS Environmental Sensing Platform, Streamflow, and Water Quality Work Plans

Budget Amount: \$368,415.00 **Estimate:** \$368,415.00 **Source of Funds:** Rates

Narrative / Description:

This item is for the renewal and modification of the cooperative USGS streamflow and water quality monitoring work plans that have been ongoing since 1989. Real-time data obtained through this cooperative agreement provides operational resiliency and historical data is used in both lake and watershed management. The total cost of the 2025 Lake Winona and Lake Maumelle monitoring plans is \$449,455.00; CAW's direct cost is \$368,415.00. The monitoring plan is divided into the following components.

Description	Overall Cost 2025	CAW Share 2025	CAW Share 2024
Streamflow Gauging Stations	175,940.00	175,940.00	172,720.00
Environmental Sensing Platform	51,935.00	26,935.00	26,935.00
Water Quality Sampling and Analyses	221,580.00	165,540.00	164,020.00
Investigative Work, Data Analysis & Program Management			10,000.00
	\$449,455.00	\$368,415.00	\$373,675.00
Total CAW Direct Costs		\$368,415.00	\$373,675.00

CAW pays 100% of the cost of streamflow monitoring. USGS directly contributes \$25,000.00 to the Environmental Sensing Platform at Lake Maumelle. USGS directly contributes \$56,040.00 to water quality monitoring. CAW provides an in-kind contribution of \$27,140.00 by providing a boat at Lake Winona.

Recommendation:

Staff recommends renewing USGS monitoring work plans.

Bid Amount: \$368,415.00

Additional Comments:

The total direct cost to CAW in 2024 was \$373,675.00. CAW's cost for the USGS Investigative Work, Data Analysis & Program Management was decreased slightly from 2024, since a summary report of historical water quality data was completed. This report is undergoing peer review and is expected to be published in 2025.

Signature: Sam Zehtaban

Title: Director of Water Production

DATE _____

AMT _____

CENTRAL ARKANSAS WATER
BOARD OF COMMISSIONERS
AGENDA ITEM

Date: February 13, 2025 **Department:** Water Production & IS **Project ID#:** _____

Item Title: Contract with Sanborn Map Company, Inc for Lake Maumelle Bathymetry

Budget Amount: \$200,000.00 **Estimate:** \$198,704.47 **Source of Funds:** Rates

Narrative / Description:

Sanborn will collect detailed bathymetric data and produce derived elevation products at a minimum, covering the central and eastern end of Lake Maumelle, starting at Grid 5 as illustrated on the provided map, and proceeding eastward. Sanborn expects to collect all Grids 1-5, including re-collecting the area closest to the dam to facilitate the investigation of a dirt slide that occurred recently, as illustrated in the map that is provided.

Recommendation:

Staff requests Board approval to exercise the second of three one-year extensions to the existing contract with Sanborn in the amount of \$198,704.47 to complete the bathymetric mapping of Lake Maumelle.

Bid Amount: \$198,704.47

Additional Comments:

In 2023, Sanborn started gathering detailed in-depth information in Lake Maumelle's western area. More data was gathered in 2024. This effort was not just about understanding how sediment settles and river channel changes but also about studying the area around the dam and intake. The insights gained have been crucial for discussions about lake management and other related activities.

Signature: Alex Harper

Title: GIS Manager



The Sanborn Map Company, Inc.
1935 Jamboree Drive, Suite 100
Colorado Springs, CO 80920
www.sanborn.com

January 31, 2025

Alex Harper
GIS Manager
Central Arkansas Water
221 E. Capitol
Little Rock, AR 72202

Re: Quotation for Continued Bathymetric Survey Data Collections and Services

Dear Mr. Harper:

On behalf of the Mapping Division of the Sanborn Map Company, Inc. (Sanborn), I am pleased to submit the attached scope of work and price quotation for continued bathymetric data collection. With this amendment, Sanborn anticipates collecting detailed bathymetric data and producing derived elevation products at a minimum covering the eastern end of Lake Maumelle. The price for this work will be \$198,704.47.

Sanborn appreciates your business, and we look forward to continuing our mutually beneficial relationship with CAW on this statement of work and in the future. If you have any questions, please contact Matt Hiland, Director of Strategic Accounts, at mhiland@sanborn.com or (512) 736-7001.

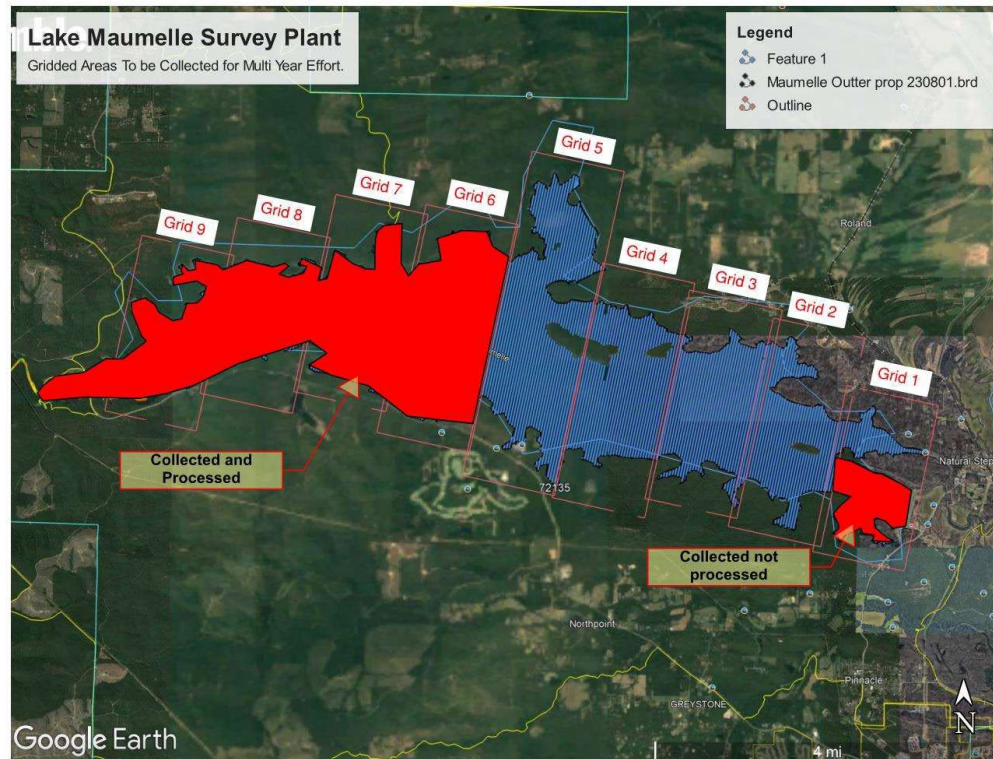
Sincerely,

A handwritten signature in black ink that reads 'John R Copple'. The signature is written in a cursive, flowing style.

John R. Copple,
Chief Executive Officer
The Sanborn Map Company, Inc.

Scope of Work:

Sanborn will collect detailed bathymetric data and produce derived elevation products at a minimum covering the eastern end of Lake Maumelle, starting at Grid 5 as illustrated in the map below, and proceeding eastward. Sanborn expects to collect all of Grids 1-5, including re-collecting the area closest to the dam to facilitate investigation of a dirt slide that occurred recently.



Mobilization and Demobilization of the Survey Vessel will consist of four (4) total days, two (2) days travel from home base to the site, and two (2) days travel from site to home base.

Two (2) days of QA/QC coordination (8-Hr Days) prior to traveling to the site. This will be GAMS calibration Bar Checks. Sanborn also plans on-site data collection from our survey vessel. The purpose is to overlay the previous data as well as Lidar data to ensure that the survey vessel is collecting data that meets the requirements for the project and there are no issues in processing and combining the data for the project.

Ten (10) days, (10-Hr Days), Ping DSP data collection for Lake Maumelle. Assumption is ½ of the time will be collecting data and ½ will be on the water travel, sound velocity casts, and QC observations for the data collection. The assumed survey speeds for data collection have been calculated at four (4) to five and one half (5.5) knots.

Sanborn will run planned lines at a 50-meter spacing for this project. This method is also called skunk striping. Our plan is to observe a maximum line spacing of 50 meters with a maximum gap between lines of 10% of the swath coverage in areas of little to no relief. This will allow us to maximize our productivity on

the project. It is not essential to collect multibeam data on areas of the lake where there is little to no relief and densify our coverage in the areas where features are detected.

Stand-by days due to weather are not calculated in this proposal. In the event we are not able to run in a 24-hour period, due to weather, we will bill ½ day crew and equipment, and per-diem for that day. Excessive wind is the greatest concern and affects our ability to collect quality data.

It may be necessary to run at night to stay on schedule. Sanborn is experienced at night-time operations and as long as we can meet the minimum requirement according to our small boat safety manual, we reserve the ability to observe data at night. Sanborn will notify CAW if this need arises.

Fuel consumption and maintenance are included in the price for this project.

Sanborn will collect and process data in Hypack and will clean the data in Qimera and Cyclone. This process will run coincidentally to the data collection process. Sanborn will deliver a calibrated processed point cloud, on datum.

The datum and control need to be established before the work commences. We will be collecting data on the chosen datum with Center Point RTX to reduce the post processing afterwards. Static data will be collected daily in the event post processing is needed in PosPac.

At the close of the project, Sanborn will develop a project report describing all our work, including means and methods. All our raw data and daily files will be delivered with all scans of our field observations logs.

Sanborn will need to coordinate with CAW to use the launch and dock at the pumping station at Lake Maumelle. We will need to utilize the shore tie on the dock at the pumping facility. CAW will ensure it is operational.

All data to be collected with RTK Tides method or similar manner to ensure accuracies are similar to RTK Tides method or be PPK post processed to meet the project standard.

Use of an Inertial Navigation System is required and shall have accuracies necessary to meet the project standard. Proof of GAMS calibration and the results from that calibration will be submitted for the project. INS data to be collected daily when the INS is in service.

Sound Velocity shall be collected, and all data shall be corrected to the specific speed of sound of the AOI daily as needed to ensure the data meets the project standard. All sound velocity probes shall have a current “less than one year old” calibration from the manufacturer.

Crossing lines shall be observed on the project to provide a statistical report on data meeting the project standard. Crossing lines shall be observed daily and overlap the previous day’s data in a manner accepted by industry standards.

Deliverables:

- Proof of Vessel survey for platforms used on project
- Bar Checks of all sensors and the calibration of the bar used on the survey
- All Daily files including the following:
 - Raw Data in Hypack format observed on the project

- Sound velocity casts used on the project and the Date, Time, and Location where the cast was made.
- Field Books and daily files on the project (digital PDF format)
- Processed Data in Hypack format for the project
- Patch Tests and GAMS calibrations conducted on the project
- INS data logging observed on the project
- A fully processed calibrated .las file to incorporate with Lidar data previously collected for the AOI
- Project report describing means and methods utilized on data collection processing calibration and deliverables. Descriptions of objects of interest.

Other Electronic Data Deliverables:

- Points
 - Shapefile (*.shp)
 - Conventional and Sweep Sonar Bathymetric Data
 - Conventional and Single Beam Sonar Bathymetric Quality Assurance Quality Control Points
 - Excel Files (*.xlsx)
 - Bathymetric Data Accuracy Assessment
- Vectors
 - Shapefiles (*.shp)
 - Topographic Breaklines

Schedule:

- The project kickoff meeting will be conducted within five (5) business days of award.
- Contracting and weather permitting, acquisition will occur in early-to-mid May, 2024.
- Data deliverables will be delivered within fifty-five (55) business days after successful acquisition.
- CAW review of data deliverables will be due within ten (10) business days of delivery.
- Revised deliverables will be due within ten (10) business days of review comments from CAW, (if necessary).
- Final CAW acceptance will be due within ten (10) business days of delivery of the revised deliverables, (if necessary).

Coordinate Reference Systems (CRS):

- UTM Zone 15 North NAD83 (2011) Epoch (2010.0000) Meters

- NAVD 88 Meters

Project Standard:

- IHO, Special Order, contained within IHO manual S-44, Edition 6.1.0 or current manual
- Object detection according to the standard 2 meters and greater may be omitted. However, objects of interest shall be reported, i.e., structures, vessels, or any other objects of interest determined by the client.
- Single beam data shall meet the standard, but 100% coverage is not required. Minimum line spacing shall not exceed 15 meters or 50 feet.
- All data shall be corrected for sound velocity.

DATE _____

AMT _____

CENTRAL ARKANSAS WATER
BOARD OF COMMISSIONERS
AGENDA ITEM

Date: February 13, 2025 **Department:** Engineering **Project ID#** 5650

Item Title: 2-Inch Galvanized Water Main Replacements Little Rock 2025 Phase 2

Budget Amount: \$2,200,000 **Estimate:** \$2,200,000 **Source of Funds:** Rates

Narrative / Description:

This project involves replacing approximately 7,410 feet of 2-inch galvanized pipe water mains in areas near Boyle Park and Rock Creek Golf Course and areas located south of Colonel Glenn Road and west of University Avenue, Little Rock. Some of the 2-inch galvanized pipes (installed between 1924 and 1965) being replaced have sections of 2-inch PVC main (totaling approximately 1,000 feet) from previous repair work. Due to water pressure limitations on PVC pipe, some of the 2-inch mains will be replaced with 6-inch ductile iron pipe in lieu of 3-inch PVC pipe. The new 6-inch mains will improve fire flows in the project areas and support nine new, additional fire hydrants that will be installed as part of this project. The 2-inch water mains to be replaced were identified by CAW's Distribution Department and the CAW asset management program as being very problematic, with high rates of leaks and breaks. Approximately one-third of the project area is expected to have rocky soil conditions. To address this, the contract bid form includes a specific pay item for trenching in rocky conditions and backfilling the new pipe with engineered fill material to protect the pipe. Much of the new main will be installed under street surfaces due to the location of underground utilities and other obstructions located within the non-paved areas of the street right-of-way. The sides of streets where the new main will be installed will be completely resurfaced in accordance with City of Little Rock ordinances. This project will eliminate two dead-end mains with looping connections to existing water mains.

The bid opening for this project was held on February 4, 2025, with Cisneros Family Construction (Jacksonville, Arkansas) submitting the low bid in the amount of \$1,893,499.26. One other bid was received. The bid results and location map are attached.

Recommendation:

Approve a construction contract with Cisneros Family Construction in the amount of \$1,893,499.26.

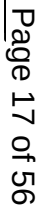
Contract Amount: \$1,893,499.26

Additional Comments:

Cisneros Family Construction is our contractor for two galvanized main replacement projects currently in progress. They have also completed several projects for CAW over the last five years. Their work has been satisfactory.

Signature: Jim Ferguson

Title: Director of Engineering



BID OPENING

2" GALVANIZED WATER MAIN REPLACEMENTS LITTLE ROCK 2025 PHASE 2

CAW PROJECT NO. 5650; JOB NO. 09351

OPEN FEBRUARY 4, 2025, 10:00 AM

APPARENT LOW BIDDER Cisneros Family Construction

BIDDER		TOTAL BID
1.	Cisneros Family Construction	\$1,893,499.26
2.	Diamond Construction Company	\$2,124,000.00
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.	CONTRACTOR'S AVERAGE	\$2,008,749.63
17.	ENGINEERS ESTIMATE	\$2,200,000

DATE _____

AMT _____

CENTRAL ARKANSAS WATER
BOARD OF COMMISSIONERS
AGENDA ITEM

Date: February 13, 2025 **Department:** Engineering **Project ID#:** 5703

Item Title: Water System Optimization

Budget Amount: \$380,000 **Estimate:** \$378,400 **Source of Funds:** Rates

Narrative / Description:

This project will identify areas for improvement and provide solutions to these areas with an optimization focus to drive efficiency and reliability enhancements to the distribution system performance. Upon completion of the hydraulic model calibration project, which has developed an updated, calibrated hydraulic model that accurately represents the existing system, the hydraulic model will be utilized as a tool for planning and decision-making to help optimize CAW's water distribution system. Tasks in this project scope include:

1. Future demand forecasting and future system modeling.
2. Hydraulic evaluations for level of service criteria, fire flow simulation, pipe criticality, pump station operating ranges, and pressure zones.
3. Water quality evaluations to evaluate storage tank turnover, water age, and chlorine residual simulations.
4. Energy evaluations to develop pump efficiency curves, cost analysis for total daily cost, and \$/MG for distribution.
5. Optimization analysis for system operation, pump station operation, real-time pump evaluation tools, and proposed system improvements.
6. Develop a final report that includes optimization results with anticipated electrical cost savings for operational changes, recommended modifications to the capital improvement plan, and a discussion of alternatives, and other recommendations.

The engineering firm, Arcadis, was selected through an RFQ process in April 2023 for a service agreement to provide hydraulic model analysis. The first phase calibrated the hydraulic model and is complete. This phase is an amendment to the original contract to provide water system optimization services. CAW staff recommends the award of the water system optimization services to Arcadis consulting engineers, a national firm with an office in Little Rock. The scope of services and fee has been developed and approved by CAW staff and Arcadis up to the amount of \$378,000.00 (hourly rate fee).

Recommendation:

Approve the award of an engineering services amendment with Arcadis in the amount of \$378,000.00.

Contract Amount: \$378,000.00

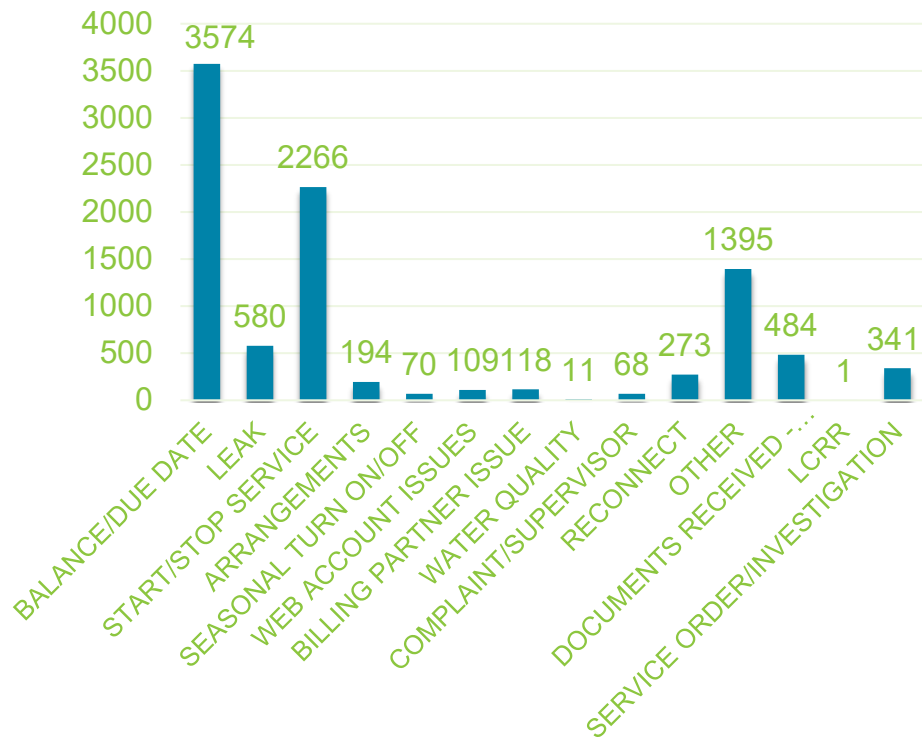
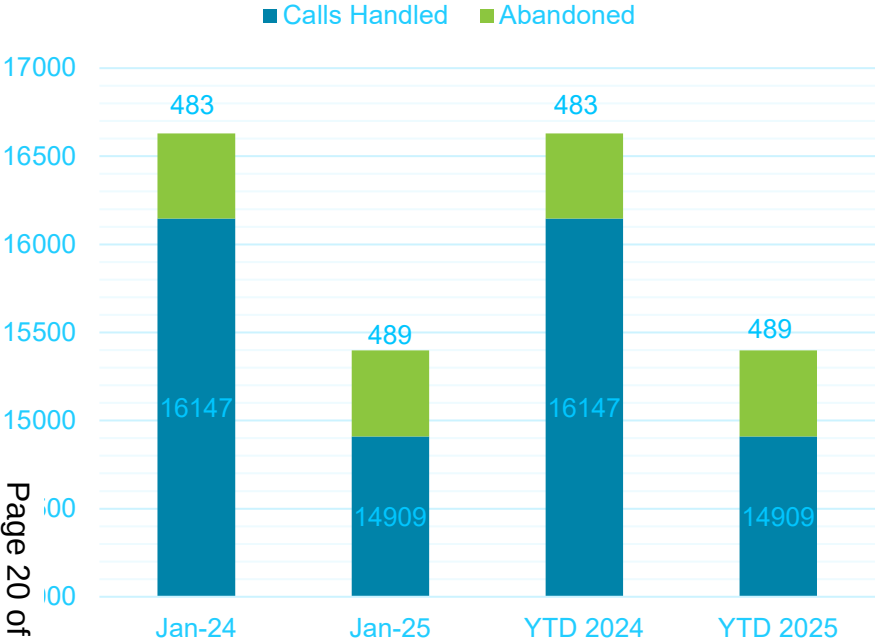
Signature: Andrew Pownall

Title: Senior Engineer

Customer Service – February 13, 2025

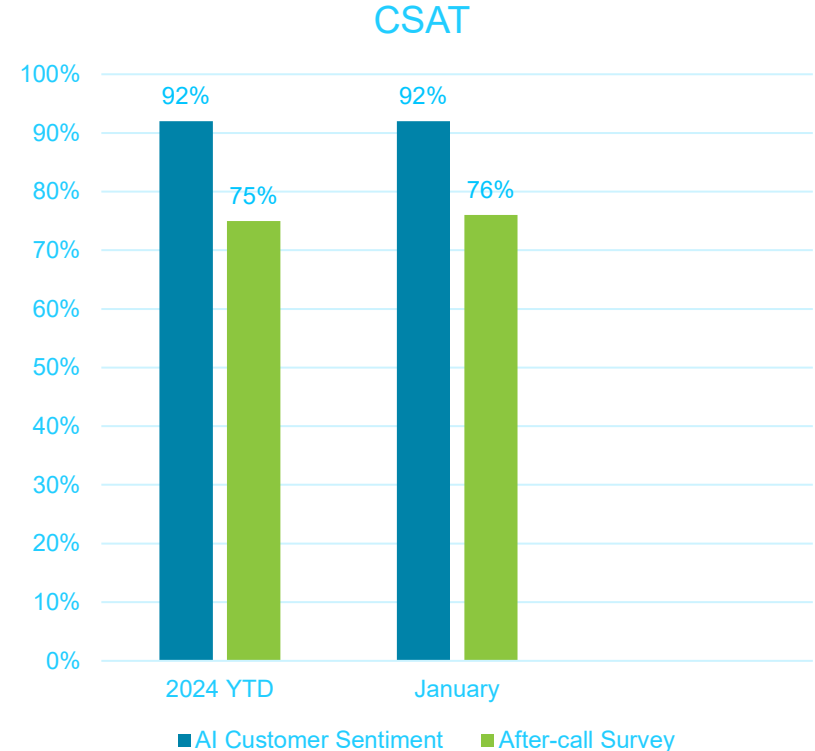


Call Volume

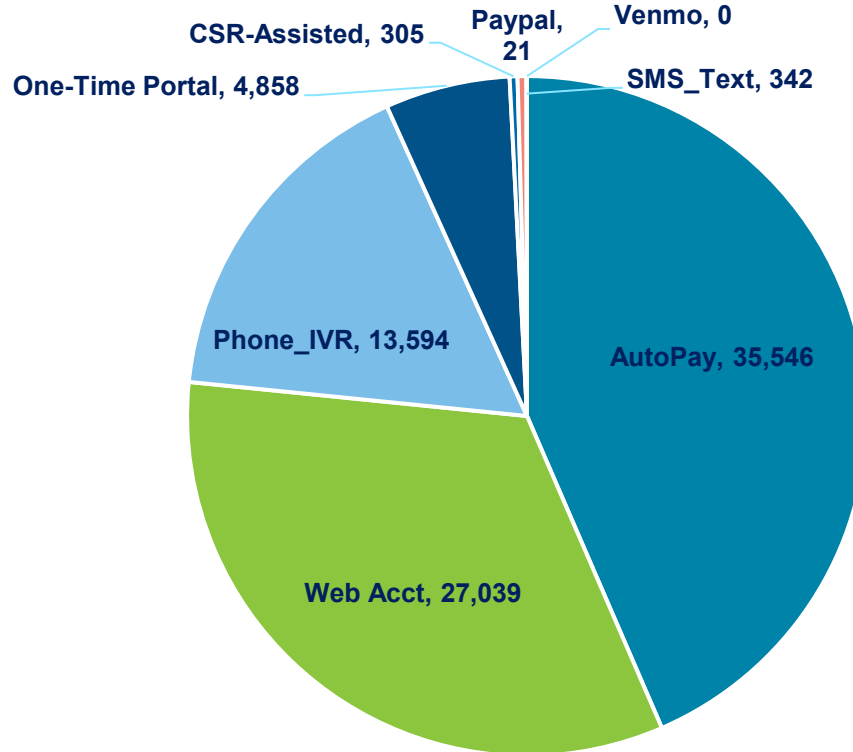


Customer Satisfaction Scores

- Dailpad uses AI technology to score each call based on customer sentiment
- AI scored 11,321 calls in January
- In February, we deployed a 1-question after-call survey: *“On a scale from 1 to 5, how satisfied were you with the service that was received today, 1 being the lowest and 5 being the highest?”*
- 395 customers answered the survey in January



Paymentus – January 2025



Customer Service – January 2025



Cashier Night Box and Mail Transactions for January:

	<u>2024</u>	<u>2025</u>
	1,135	985
Year to Date	1,135	985

Shutoffs completed January 2025	Pending Shutoffs
326	5,278

Promise Pay Outreach Efforts-LIHWAP (Low Income Housing Water Assistance Program) *July 1,2023 Promise Pay’s efforts will pivot to pay arrangements.

Total LIHWAP Assistance to date:	\$5,524,078.05
Total Promise Pay Arrangements	4,209 active; 8,990 completed Total amount collected: \$3,323,316

Customer Service Field – January 2025



Total Number of Meters Read	Accuracy Rate
118,641	99.84%

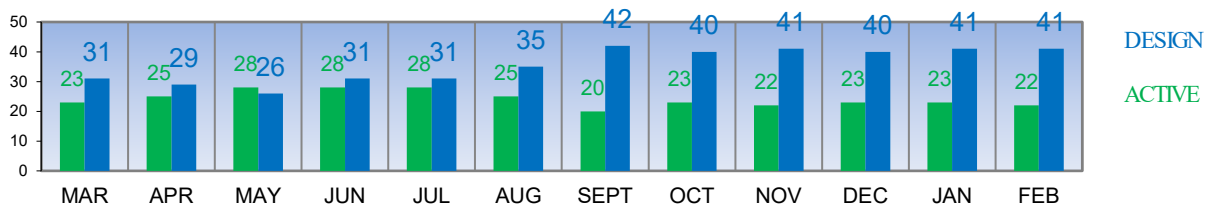
Total Number of Service Orders Completed	Accuracy Rate
11,345	99.85%

Engineering Highlights February 13, 2025

❖ 2025 CURRENT

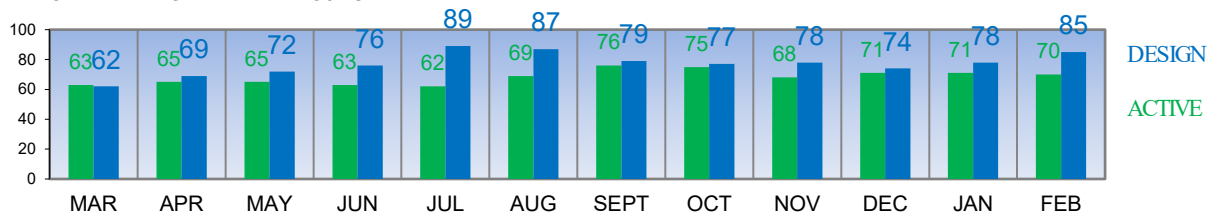
- The Engineering Department currently has 41 **capital improvement** projects in **design** and 22 **active** projects (under construction).

CAPITAL IMPROVEMENT PROJECTS



- The Engineering Department currently has 85 **developer-initiated** improvement / extension projects in **design** (under review) and 70 **active** projects (under construction).

DEVELOPER IMPROVEMENT PROJECTS



❖ PROJECT HIGHLIGHT - this month

- Elevated Tank No. 17A (Chenal Heights Dr, Little Rock) Interior and Exterior Painting – Project #5498**

This project involves painting the entire interior and exterior surfaces of Tank No. 17A, a 250,000-gallon welded steel, single pedestal, spheroid elevated tank located at 50 Chenal Heights Drive, Little Rock. This tank was last painted in 1990. The new painting work is being performed by USG Water Solutions (Perry, GA). In addition to the painting, the project also includes work to upgrade and/or replace certain appurtenances of the tank to meet current Arkansas Department of Health requirements. The CAW Board of Commissioners approved this project at the August 2024 commission meeting with a project cost of \$361,200.00. The contractor started work on the project mid-November 2024. As of this date the work is approximately 35% complete. The tank upgrade work started with the painting of the interior of the bowl, which is complete as of this date. The painting work involves removal of the old paint system primarily by “sand blasting” and then applying the new paint system. The tank appurtenances being upgraded or replaced include the roof access hatch, roof vents, over-flow outlet assembly, and tank water sample tap for monitoring water quality. Relocating cellular communication facilities located on the tank requires significant engineering and planning by the cell carrier companies. Additionally, CAW met on site with all three cell carriers to plan scheduling and locations of the temporary facilities. Signal interference and layout of ground cable runs were important issues that were discussed. Before the exterior painting work starts, a large curtain system will be erected around the tank to contain blasting media and paint over-spray, similar to the containment curtain recently utilized for the Tank No. 2 painting project. The contract time for this project is 150 days and the work is currently on

schedule. There have been no customer complaints to date. The work included in the USG Water Solutions contract also includes painting and upgrading Tank No. 22 (a 1,000,000-gallon welded steel elevated tank located in the Indian Hills area of North Little Rock). The Tank No. 22 work is approximately 50% complete.



Tank No. 17A – 0.25 MG elevated steel
50 Chenal Heights Dr, Little Rock



AT&T facilities crowding the tank roof hatch. After the painting work, AT&T will install their antennas on the existing antenna corral which is located away from the roof access hatch.

T-Mobile, Verizon, and AT&T facilities on temporary elevated structures at the tank site. Ground storage Tank No. 16B is seen in the photo. Tank No. 16B is located next door to elevated Tank No. 17A.





Section of the bottom of the tank bowl interior before surface preparation for the new paint system. Note the surface staining and sediment.

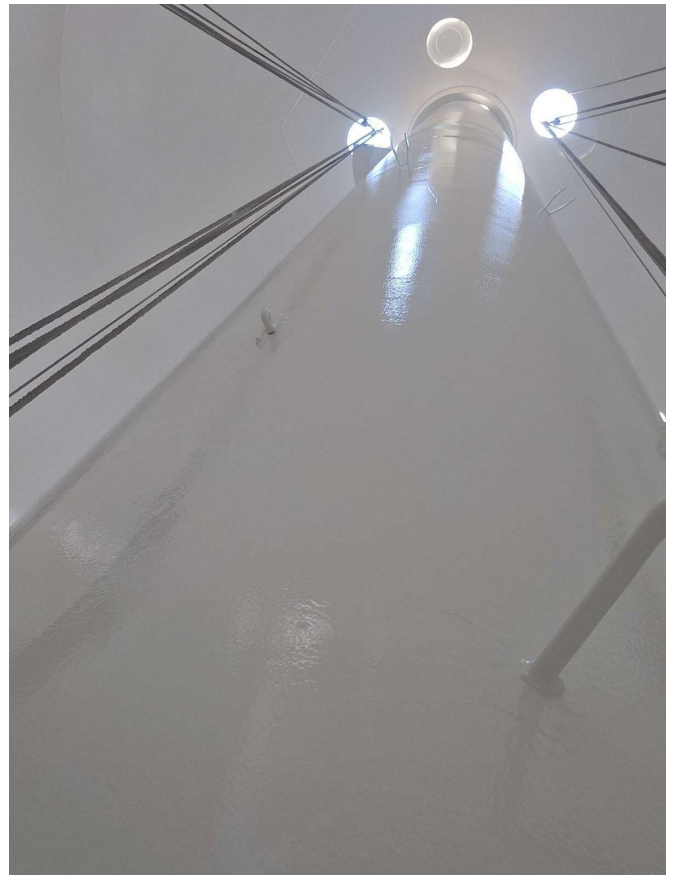


Looking up the access ladder that is located inside the pedestal that supports the tank bowl. There are three intermediate safety platforms along the 163-foot height of the tank. All of these interior surfaces will be prepared and painted.



Primer coat work being done on the inside bowl of the tank. The primer coat is spray applied, but the welded seams get an extra brush coat. To the left is the roof access opening. To the right is the ventilator opening.

The tank bowl interior after application of the finish coat.





A new steel tank vent base being welded to the roof of the tank for a new tank vent assembly.



A new tank vent assembly that will be installed on the roof of the tank (on the steel flange shown in the photograph above).



An air compressor and “sand blasting” equipment on site. Compressed air is used by the “sand blasting” equipment and paint spray equipment.

The hopper for the “sand blasting” media. Note the roll of fabric below it – fabric is placed around the base of the tank to collect “sand blasting” debris and paint over-spray. Sand blasting media will be cleaned and recycled for multiple uses.



Distribution – February 13, 2025

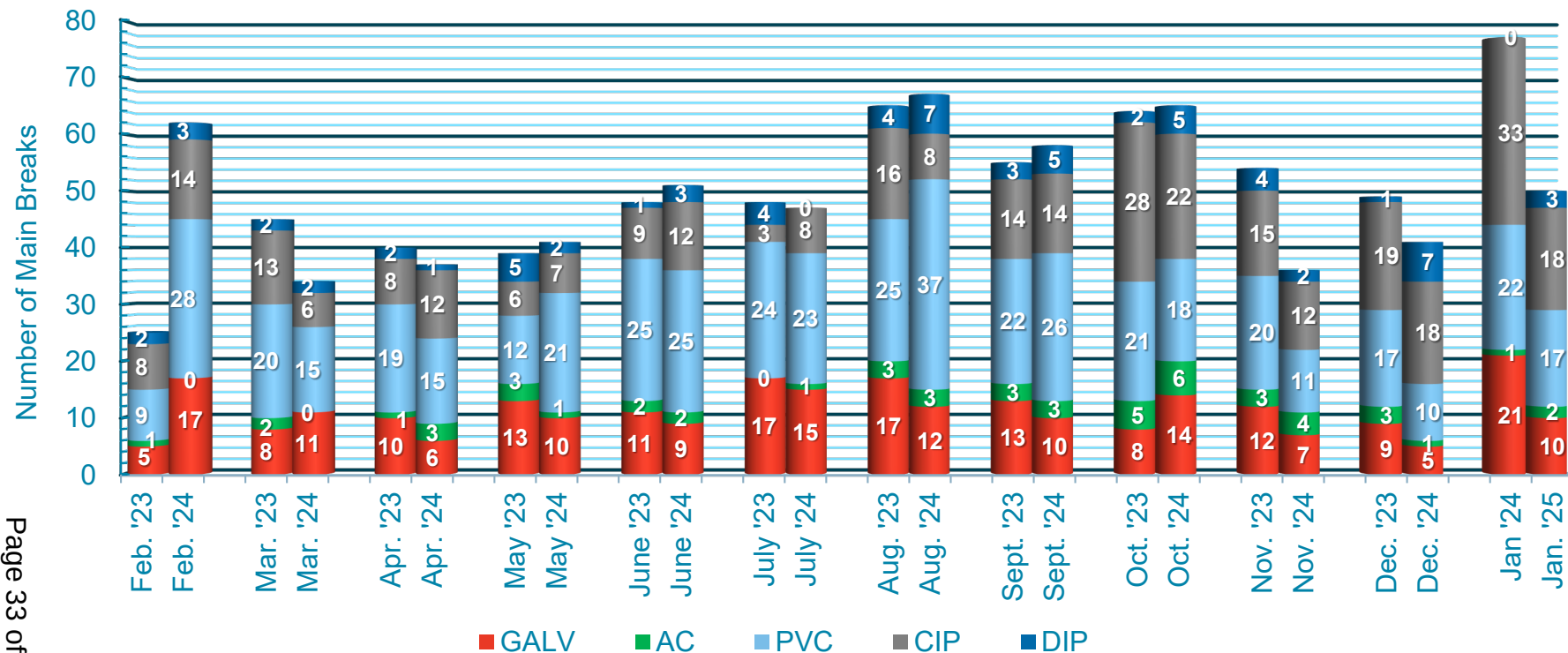
<u>Year</u>	Main Breaks		Galvanized	
	Jan.	YTD	Jan.	YTD
2025	58*	58	10	10
2024	78	78	21	21
2023	58	58	13	13
2022	59	59	17	17
2021	58	58	11	11
2020	51	51	17	17

Jan. Main Break Summary

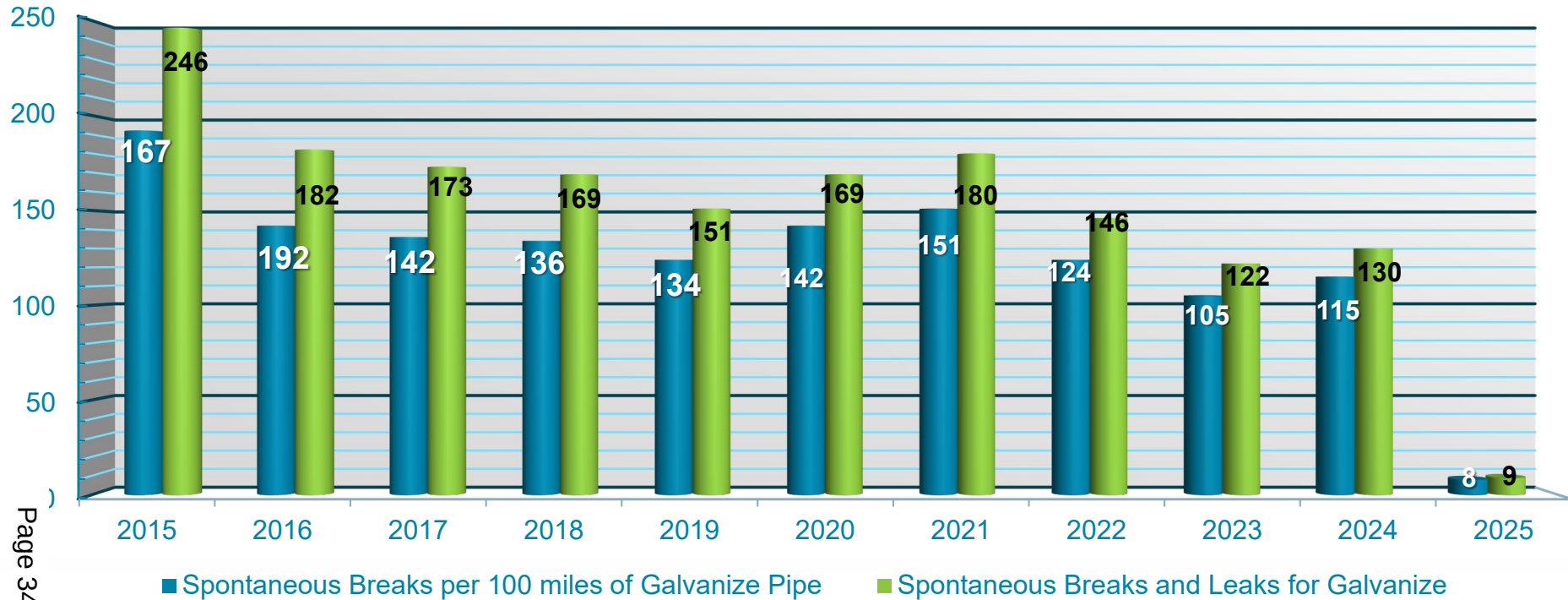
- ☐ 2 – 2" CI
- ☐ 10 – 2" Galv. (1 BBC)
- ☐ 12 – 2" PVC (7 BBC)
- ☐ 2 – 3" PVC (1 BBC)
- ☐ 11 – 6" CI
- ☐ 2 – 6" PVC
- ☐ 1 – 8" AC
- ☐ 3 – 8" CI
- ☐ 2 – 8" DI
- ☐ 1 – 10" PVC
- ☐ 1 – 12" AC
- ☐ 1 – 12" CI
- ☐ 1 – 12" DI
- ☐ 1 – 24" CI

<u>Planned / Unplanned Shutdowns (Jan.)</u>	Broke by Contractor	Customer Affected
22 – Planned affecting 301 customers	9	113
3 – Unplanned affecting 36 customers	0	0
2 – Shutdowns not required	0	
8 – Pending repairs		

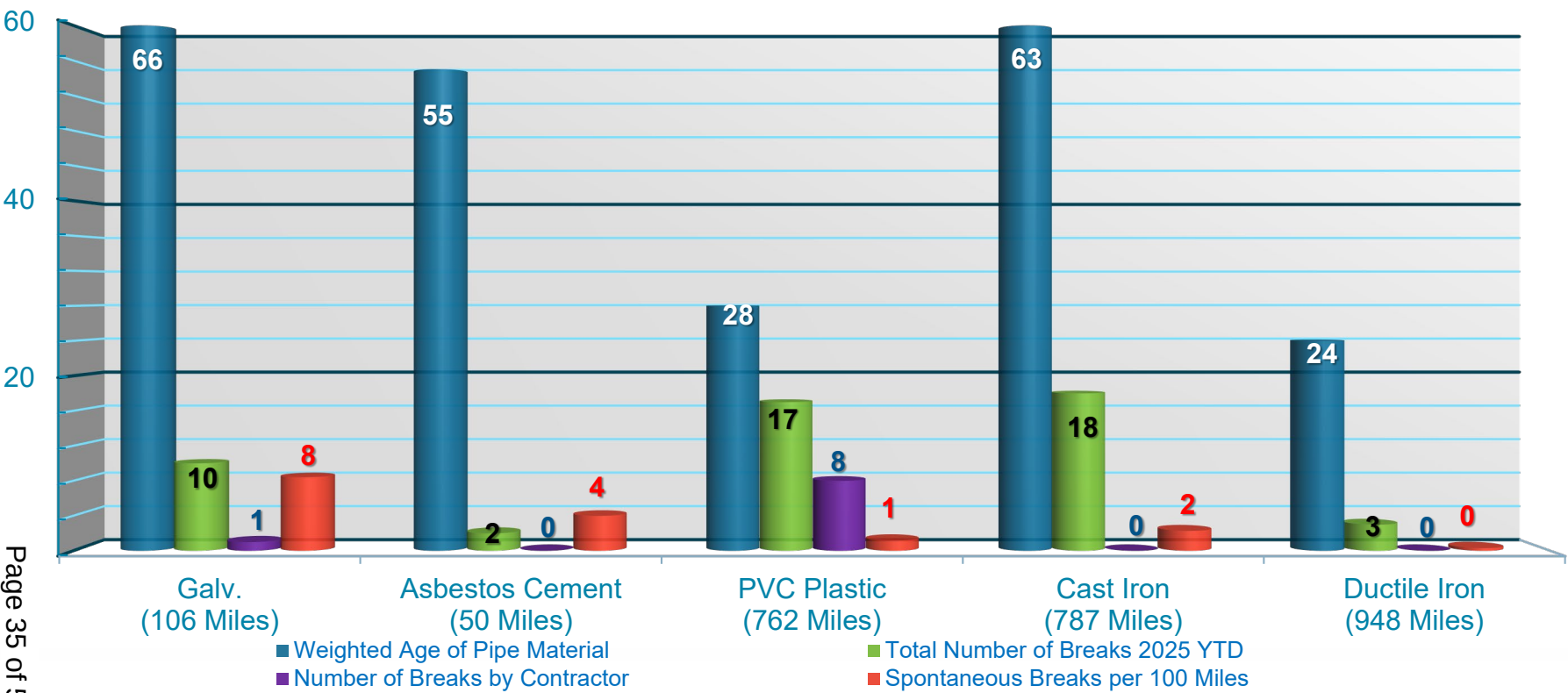
Main Break Comparison



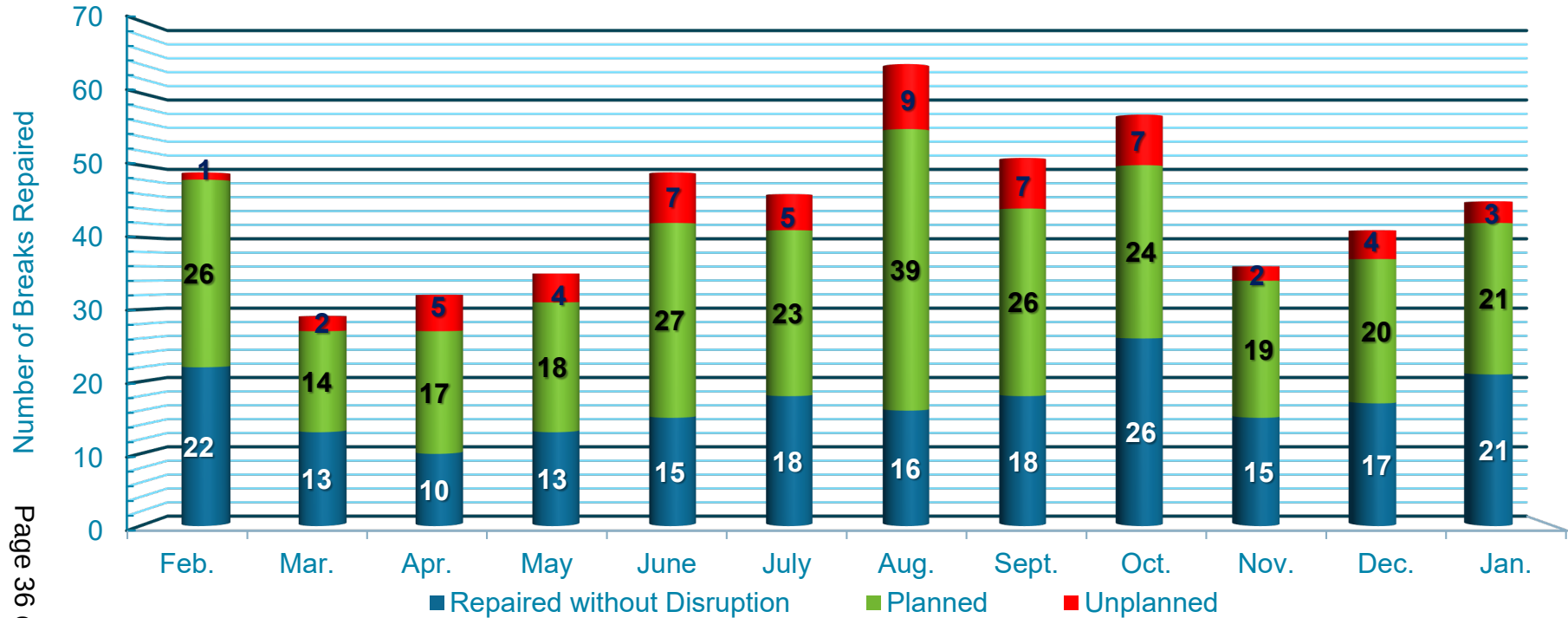
Spontaneous Breaks on Galvanize



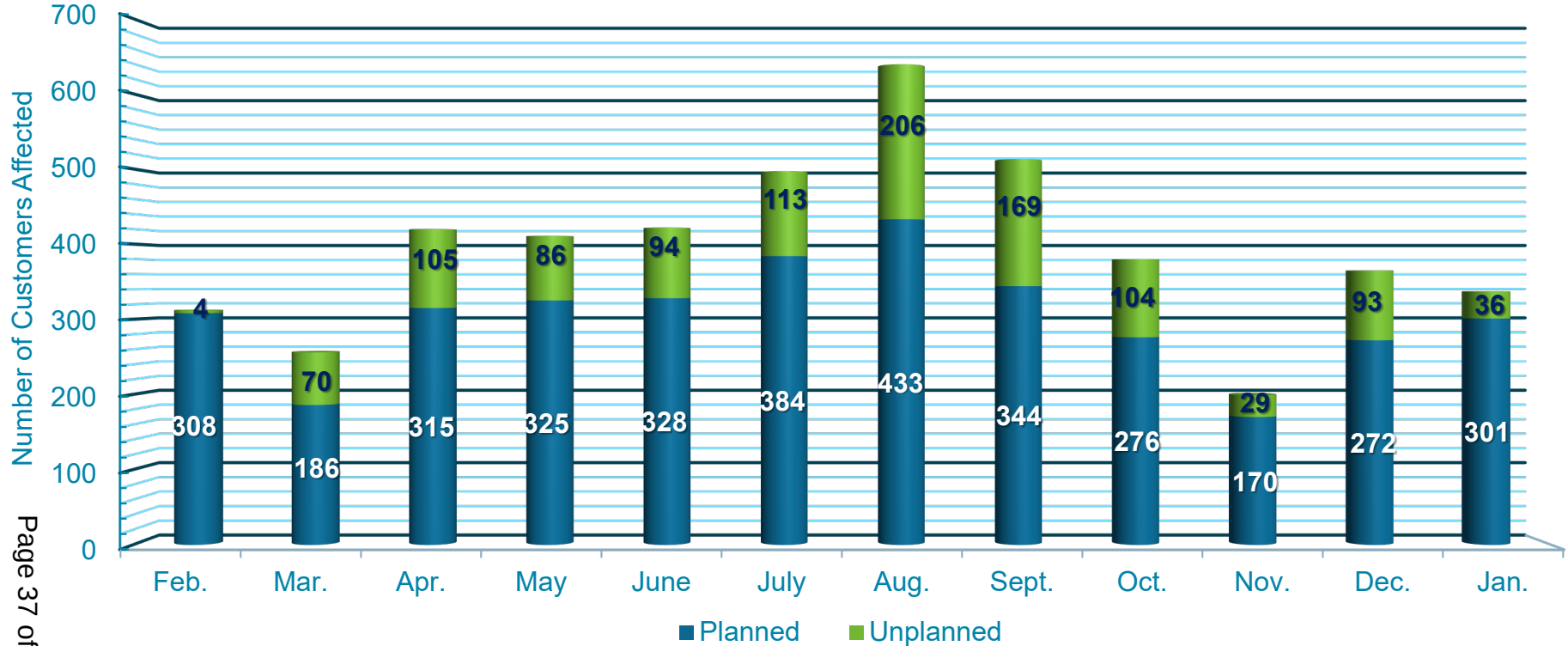
2025 YTD Breaks by Material



Planned vs. Unplanned Outage



Customer Outages



Distribution Department

	2025		2024	
<u>New Services</u>	Jan.	YTD	Jan.	YTD
1" Services:	25	25	51	51
1" Stubs:	0	0	2	2
Bullheads:	4	4	7	7
<u>Replaced Services</u>	33	33	40	40

			Jan.	YTD
Work Orders:			660	660
Meter Change-Out Program:			462 Dist.	746 w/CS
Un-Locatable Valves:			0	0
Remaining:	>12"	0		
	4"-10"	81		
	<3"	182		
Valve Inspections:				
14" & Smaller			41	218
16" & Larger			177	
Fire Hydrants Inspections:			25	25
Pipe Installed/Replaced:			560'	560'



New Meter Installs

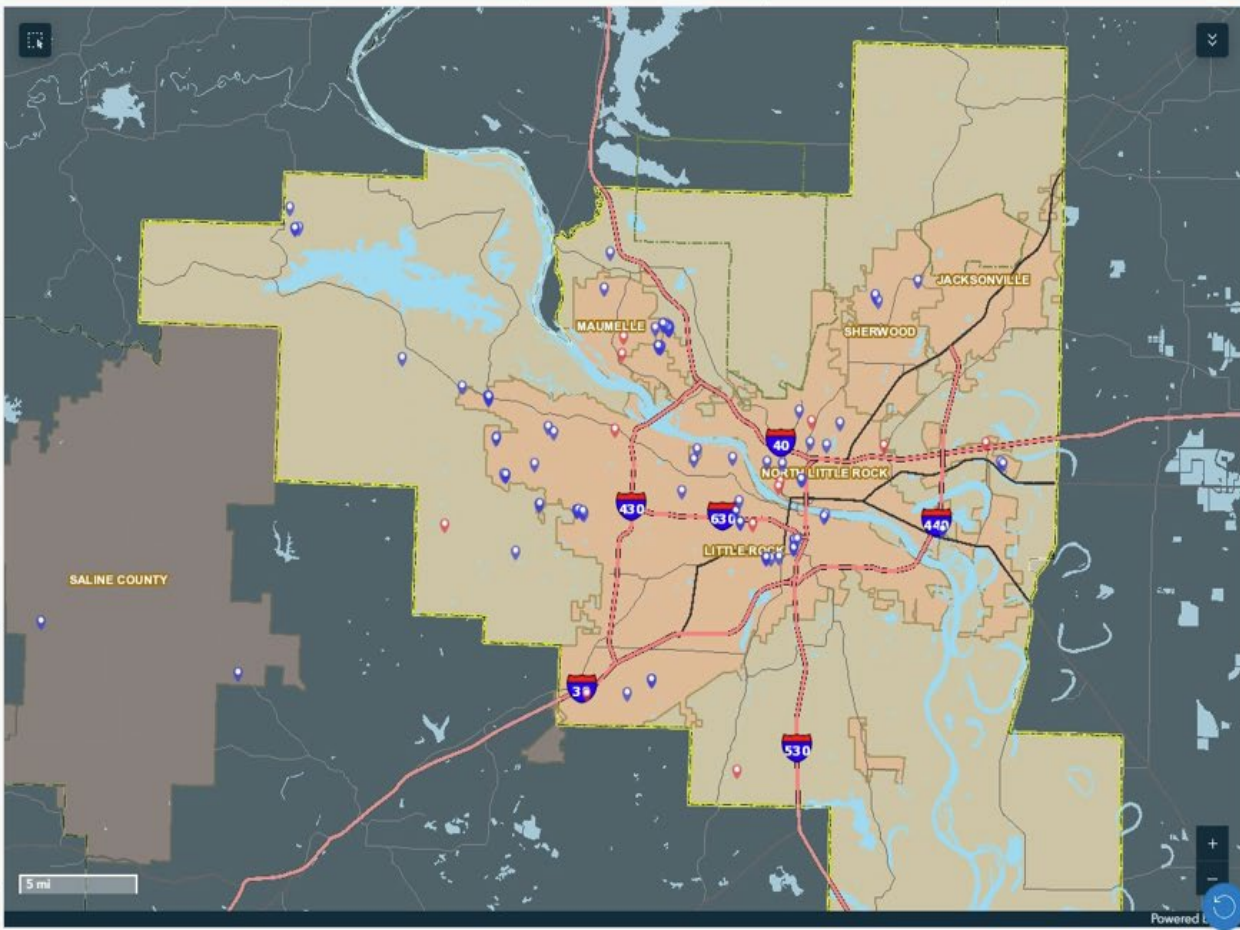
106

Installs Previous Month

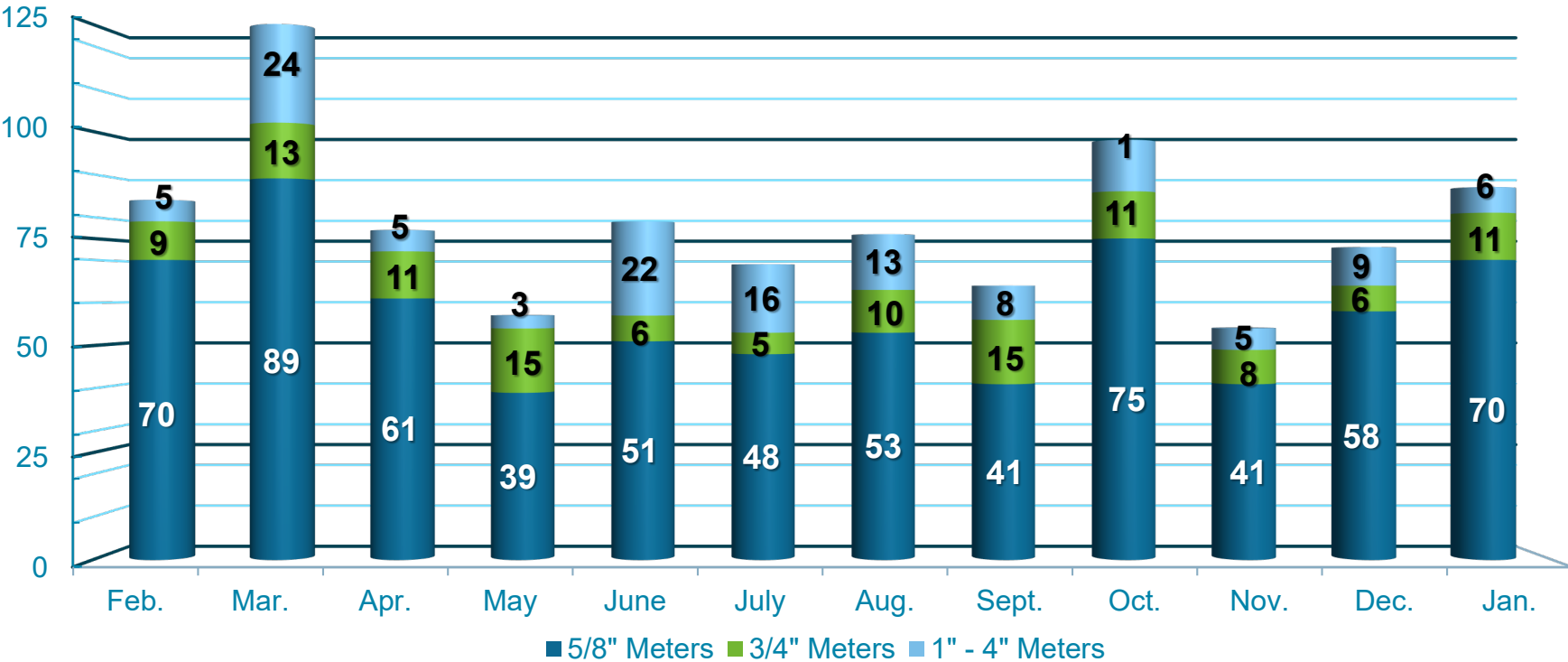
89

Installs Year To Date

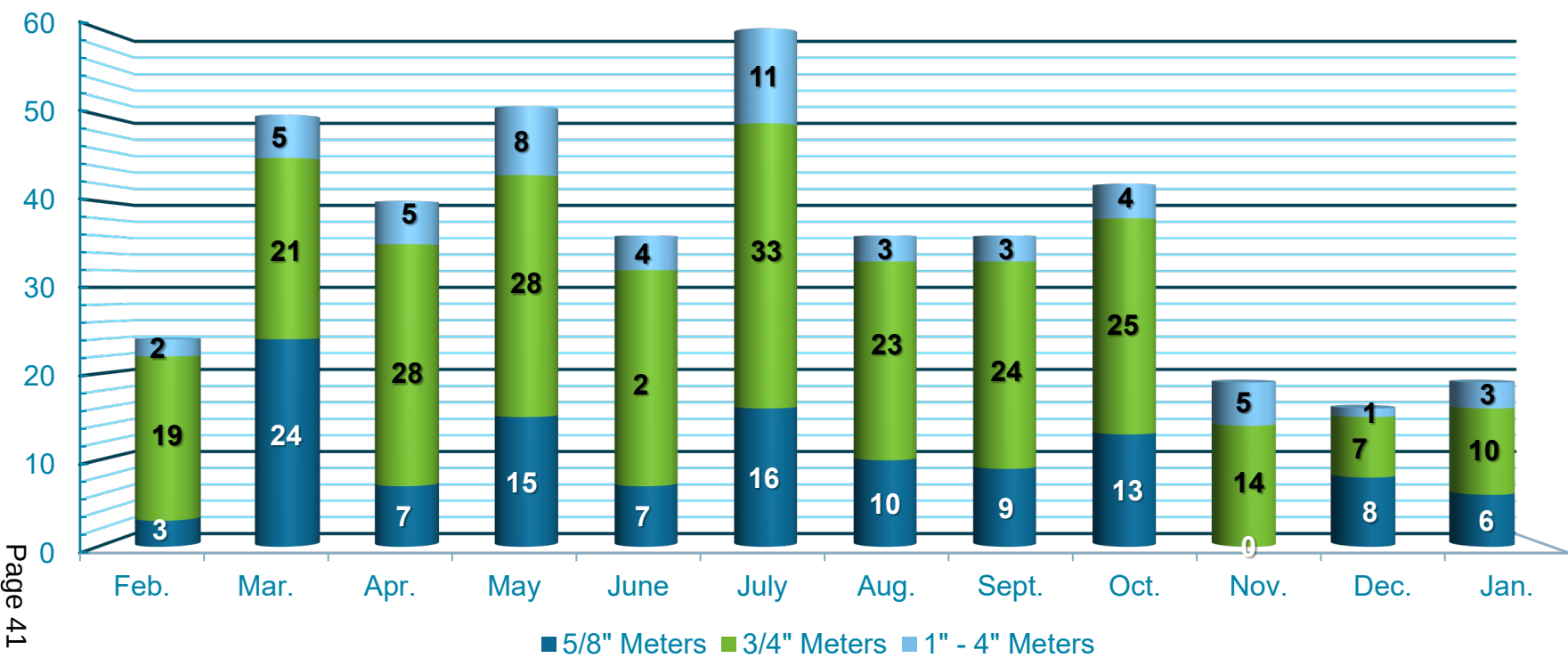
106



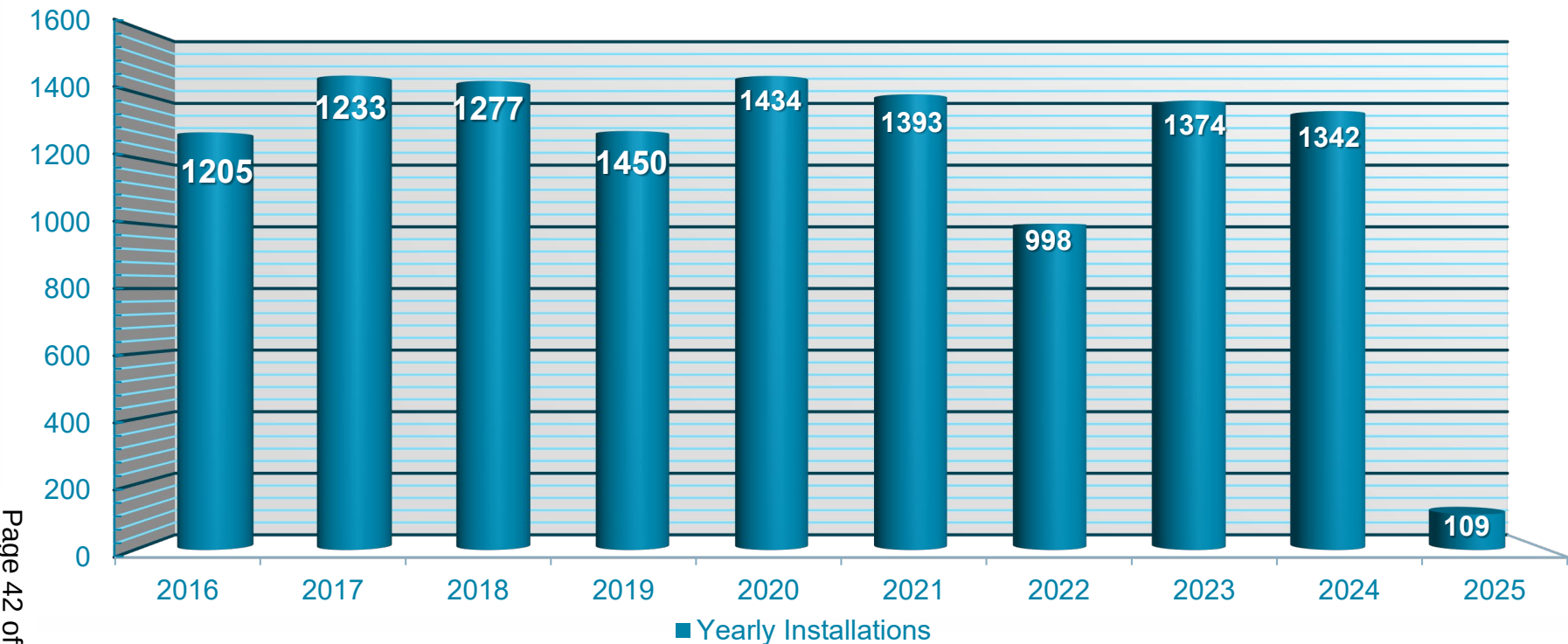
2025 YTD Domestic Service Meter Installations



2025 YTD Irrigation Service Meter Installations



Meter Installations by Year



Water Production – February 13, 2025

January Total Consumption

<u>2025</u>	<u>2024</u>
2.0 BG	2.2 BG

YTD to Date Consumption

<u>2025</u>	<u>2024</u>
2.0 BG	2.2 BG

Rainfall

<u>January</u>	<u>YTD</u>
3.53"	3.53"

Lake Update

Lake Winona	0.81 feet above spillway
Lake Maumelle	6.99 feet below spillway

Ozark Treatment Plant:

- ❑ Repaired lime feeder #1 discharge line
- ❑ Repaired basement lighting

Wilson Treatment Plant:

- ❑ Repaired disconnected feed line on basin 4
- ❑ Repaired flocculator 2A-2 mixer shaft
- ❑ Installed LED lighting in maintenance building shop
- ❑ Replaced basin #2 sludge scraper motor

Paron System and Treatment Plant:

- ❑ Troubleshoot and repaired alarming chlorine sensor

Lake Maumelle and Lake Winona Pump:

- ❑ Replaced sample pump

Distribution System:

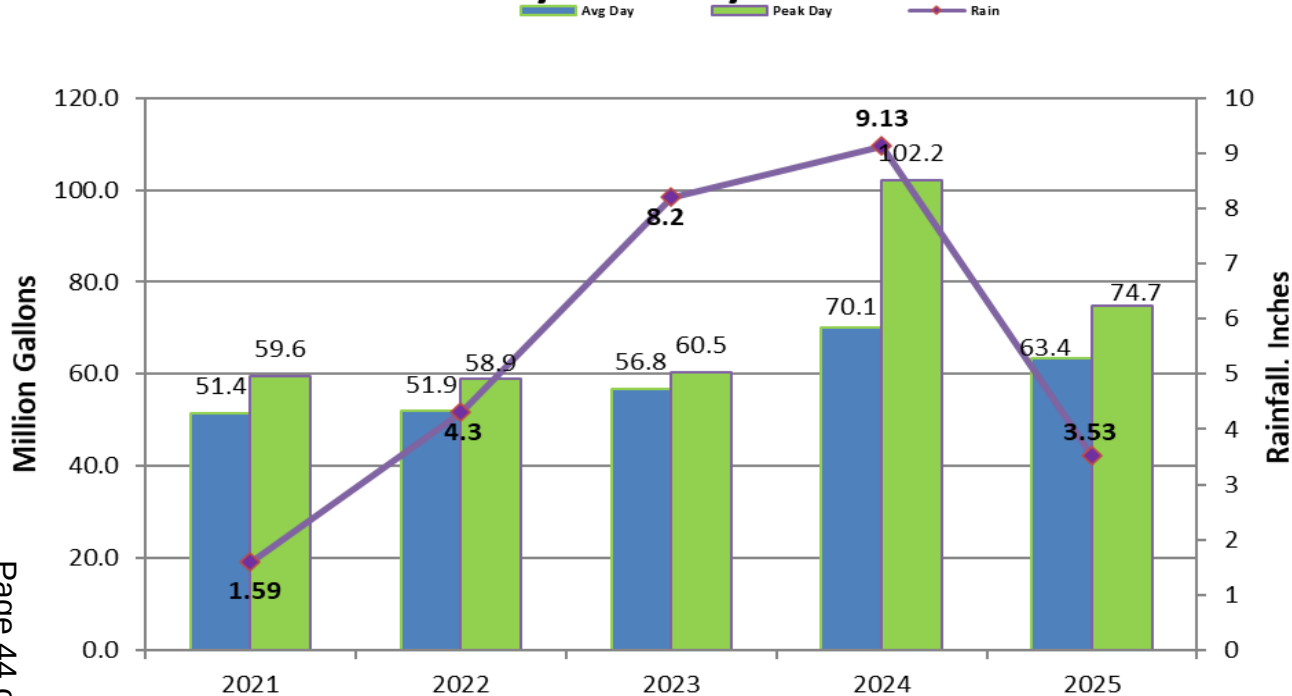
- ❑ Repaired Main Street meter
- ❑ Replaced transducer at I-430 meter
- ❑ Replaced PLC equipment for Asher Bryant meter

Wastewater:

- ❑ Wrightsville construction continues

Average/Peak Monthly Production

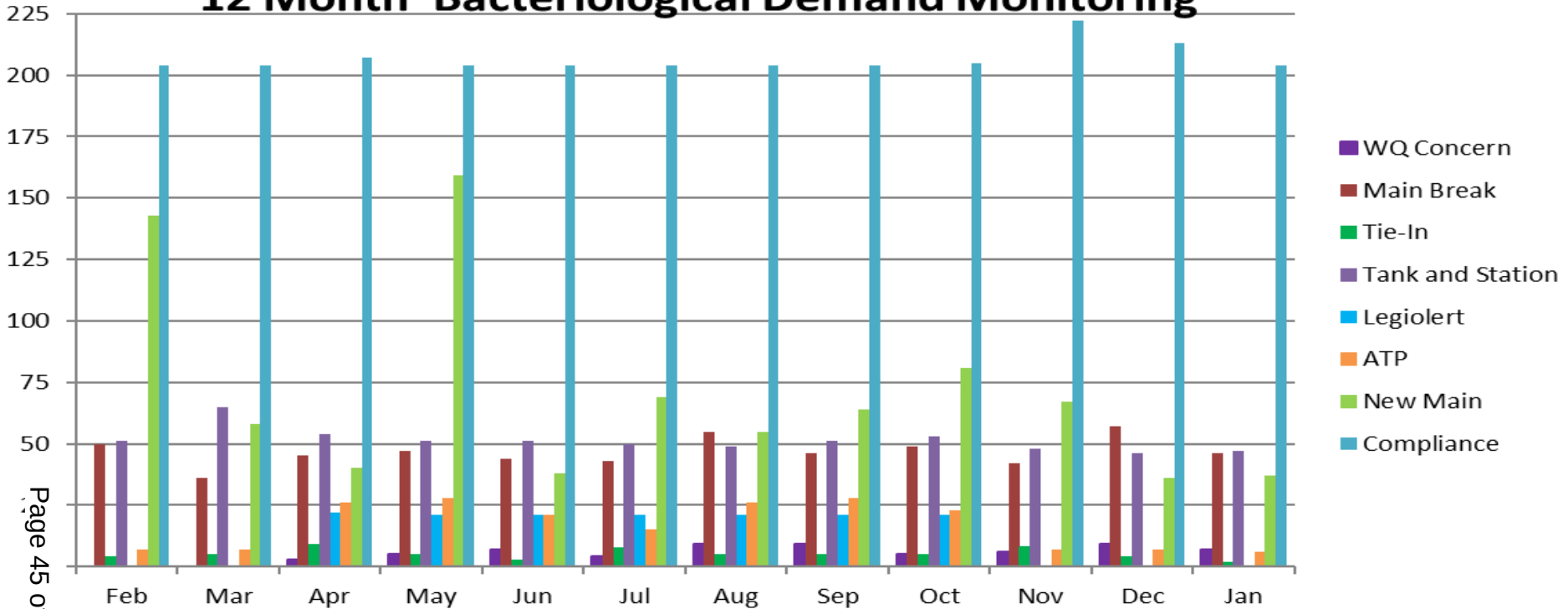
January Monthly Production



**January
Previous 5
Year Avg. Day
Consumption:
55.8 MGD**

Monthly Bacteriological Report

12 Month Bacteriological Demand Monitoring



Water Quality Sampling

❑ Total Coliform Rule Sampling

❑ # Samples collected CAW	215
❑ # Total Coliform Positive samples CAW	0
❑ # Resamples collected CAW	0
❑ # Samples collected Wye	3
❑ # Total Coliform Positive samples Wye	0
❑ # Samples collected Paron	3
❑ # Total Coliform Positive samples Paron	0
❑ # Samples collected Perla	3
❑ # Total Coliform Positive samples Perla	0

❑ New Main Sampling

❑ Projects	7
❑ Samples collected	37

❑ Disinfection By-Product Sampling

❑ # Sites monitored CAW	14
❑ # Sites monitored Wye	0
❑ # Sites monitored Paron	0

❑ January Customer Concern Summary

❑ Total # calls received	45
❑ # Samples collected	7
❑ # Positive samples	0

❑ Customer Concern Samples by Category

❑ Health	2
❑ Lead	
❑ Particles/biofilm	2
❑ Discoloration	3
❑ Taste/odor	
❑ Chlorine	
❑ Other/general inquiry	

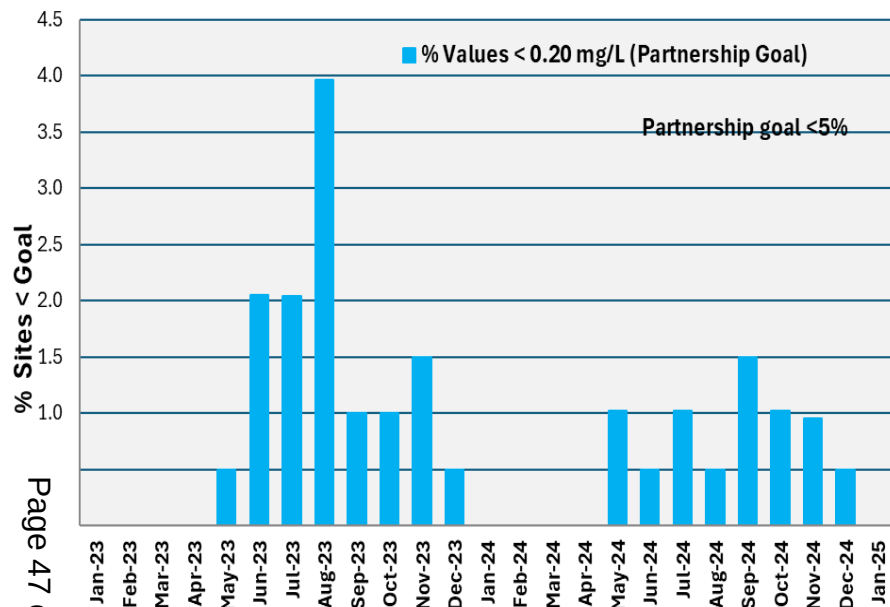
❑ Special Sampling

❑ Perla system monitoring	3
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Distribution System Chlorine Residuals

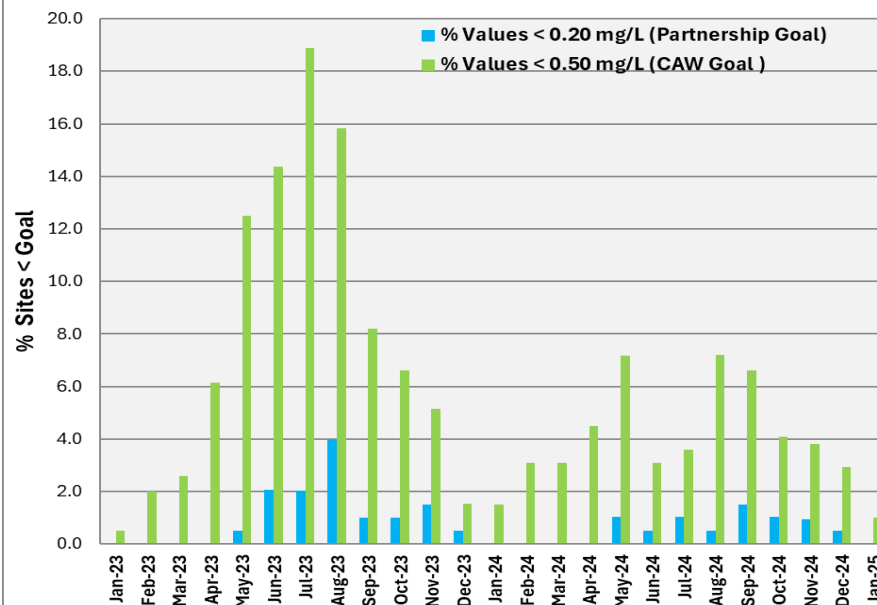
Partnership for Safe Water Goal

Percent of TCR Sample Site Free Chlorine Residual Values < Goal

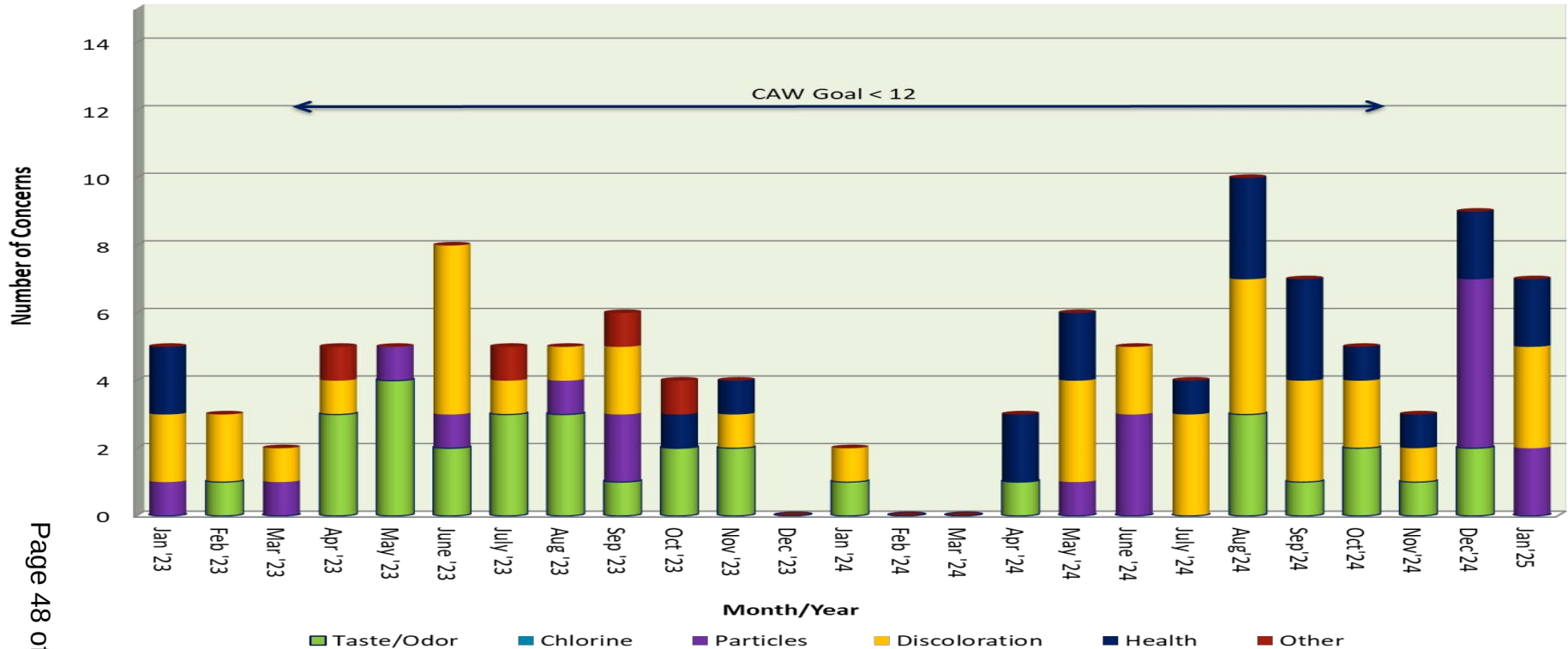


CAW Internal Goal

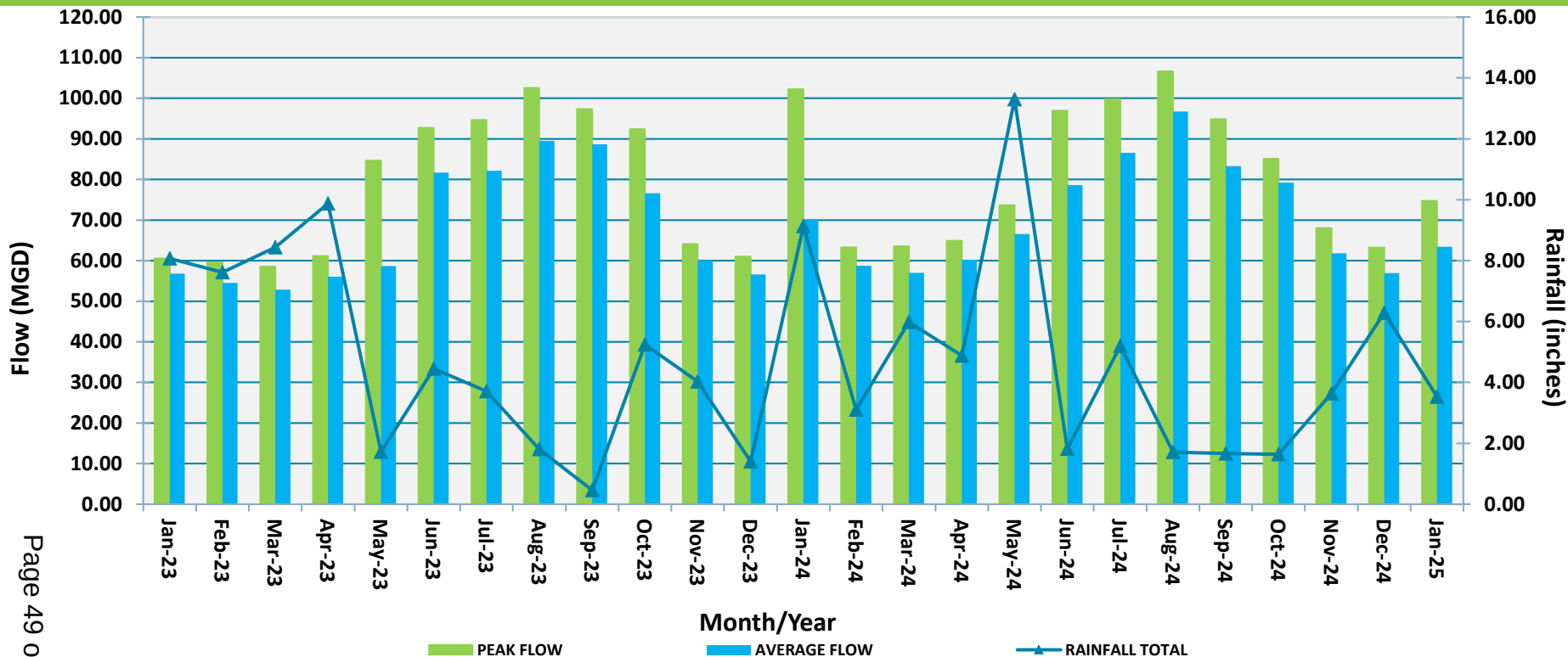
Percent TCR Sample Site Free Chlorine Residual Values < Goals



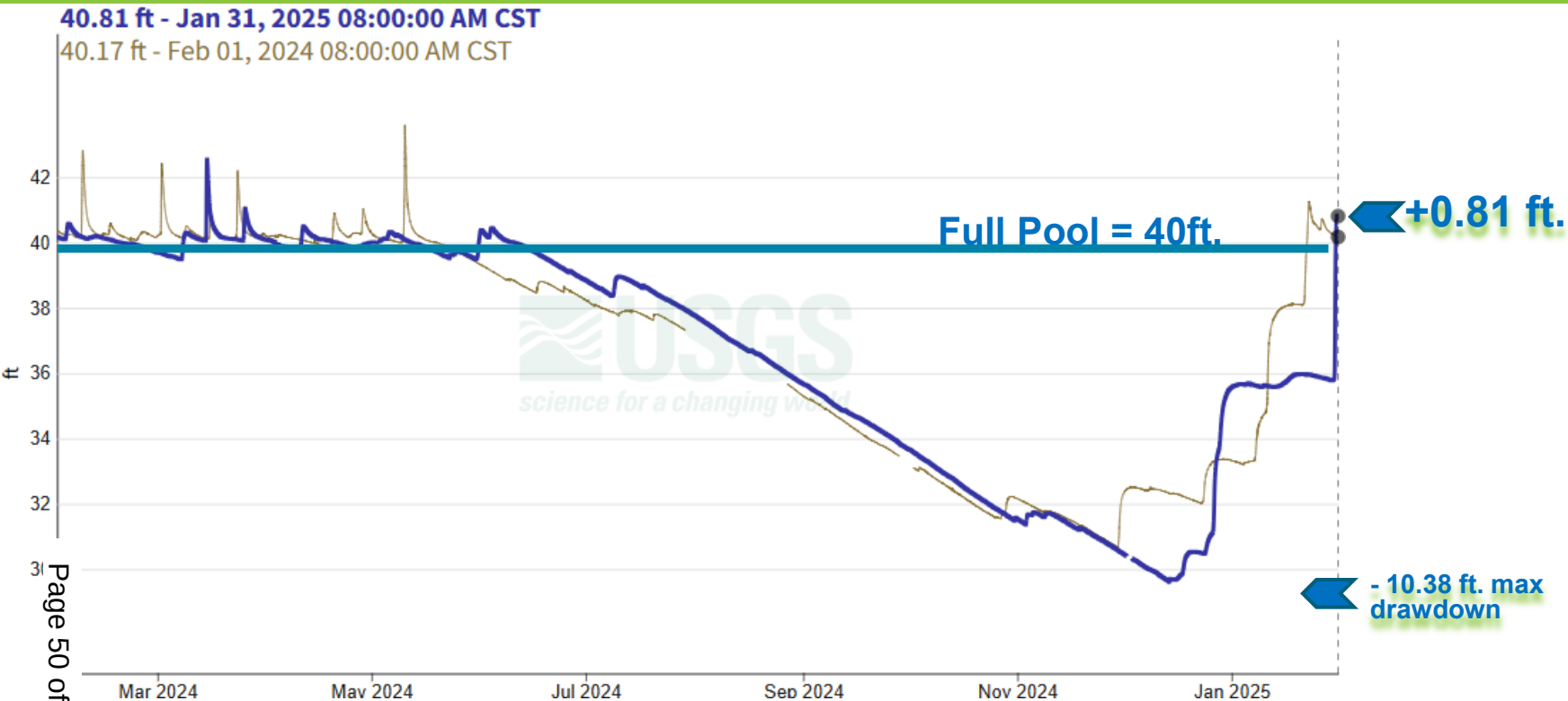
Customer Water Quality Concerns



Monthly Peak & Average Day Flow vs Rainfall



Lake Level: Winona



Lake Level: Maumelle

83.01 ft - Jan 31, 2025 08:00:00 AM CST

86.25 ft - Feb 01, 2024 08:15:00 AM CST

Full Pool = 90ft.



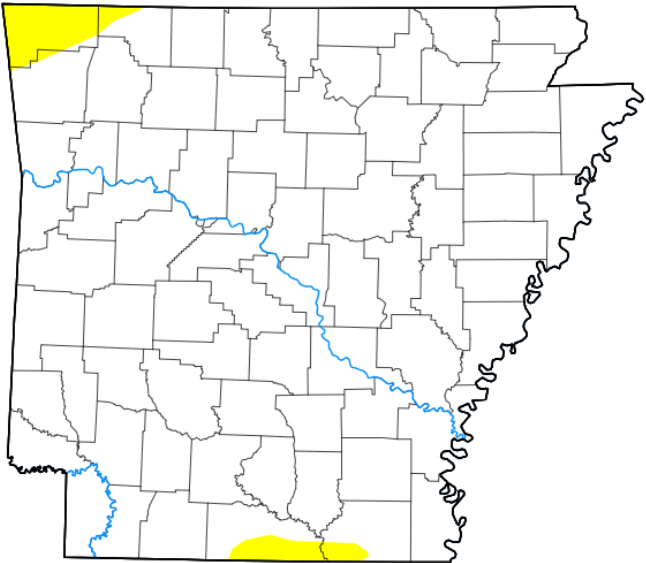
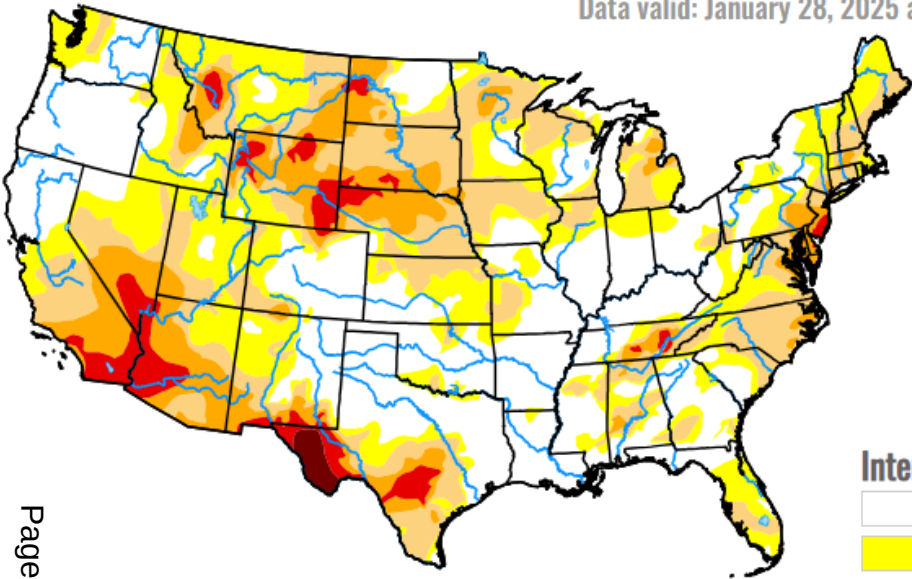
◀ - 6.99 ft.

Drought Monitor

Contiguous U.S.

Map released: Thurs. January 30, 2025

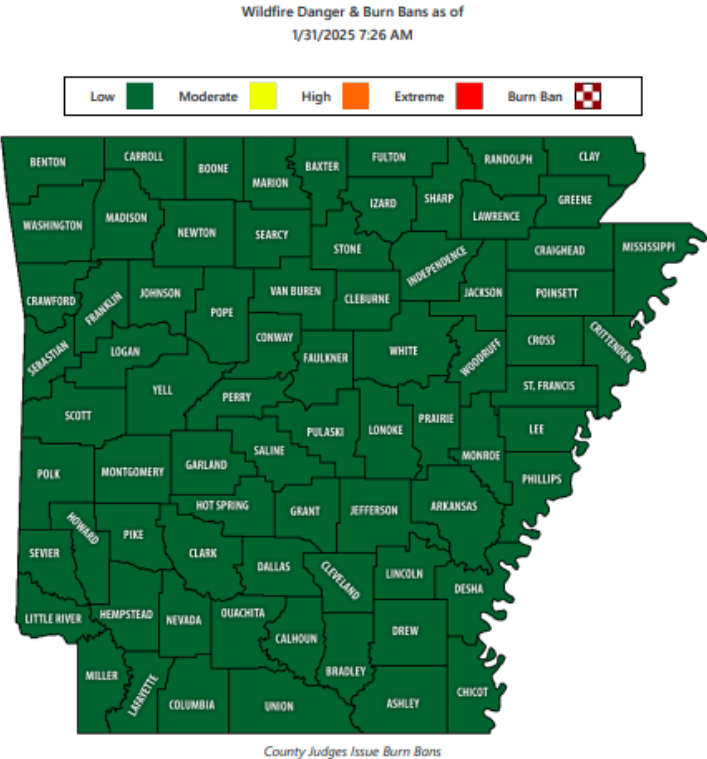
Data valid: January 28, 2025 at 7 a.m. EST



Intensity

	None		D3 (Extreme Drought)
	D0 (Abnormally Dry)		D4 (Exceptional Drought)
	D1 (Moderate Drought)		No Data
	D2 (Severe Drought)		

Wildfire Danger Status



Watershed – February 13, 2025



Key Metrics for January

LAND MANAGEMENT

Fireline Inspections	1 unit
Historic Dump Sites Cleaned	6 Sites
Tires Removed from Watershed	42 tires

MONITORING & INSPECTIONS

206 Maumelle River Restoration	2
Water Quality Monitoring	4

February Focus & Agenda Items

- **AGENDA:** USGS Contract and Maumelle Pinnacles Connector Trail

- **FOCUS:** Boundary Marking, Prescribed burning preparation, and historic dump site cleanups

Major Activities for January

Watershed Accomplishments & Notables

- ❑ ***Bryan Rupar has been promoted to Watershed Protection Manager!***
- ❑ Watershed staff reorganized the Lake House office to accommodate additional staff including Water Resources Specialists & Water Resources Technician
- ❑ 6 historic dumpsites were cleaned which included numerous large pieces of styrofoam, tires, and metal
- ❑ Watershed and Engineering Staff met with Crafton Tull to kickoff the Watershed Office at the WestRock Landing design
- ❑ CAW Website has new Trails Maps available around Lake Maumelle

CAW-Internal Partnerships/Projects

- ❑ Bryan and Ben met with other Water Production staff to continue emergency action plan discussion and planning
- ❑ Bryan attended the first CAW-U Leadership course
- ❑ Jeff and Ben responded to an accident that resulted in a minor spill within the Maumelle Watershed along Hwy 300 that has been fully cleaned up

External Partnerships/Projects

- ❑ Bryan and Ben meet with the Maumelle Pinnacle Stakeholder group
- ❑ Ben presented potential projects to the UCA Environmental Practicum Class and Jeff met with students on site to evaluate reforestation efforts on CAW land
- ❑ Ben met with Arkansas Forests and Drinking Water Collaborative on site to discuss logistics of the upcoming CommuniTree Project
- ❑ Bryan and Brad met with USGS to discuss Water Quality monitoring contracts and needs
- ❑ Bryan and Ben met with WRI and the Walton Foundation to explore potential partnerships

Industry & Community Contributions

- ❑ Brad provided water quality data to new faculty at UALR working to better understand phosphorus dynamics in Lake Maumelle

Watershed Highlights

- Watershed staff tackled numerous historic dump sites from Styrofoam to tires to decades old household trash
- An exposed sunken sailboat was removed from Lake Maumelle during the draw down



Watershed Highlights



- Upper Left-UCA Students at one of the sites thinking about questions to ask about reforestation-Photo by Jeff Reed
- Upper Right-Drone photo of the low water crossing to be removed this year during the 206 Maumelle River Restoration Project-Photo by Jeff Reed
- Lake Winona during Water Quality Sampling by Brad Meredith and Nash Fortson-Photo by Nash Fortson

